

Customer Rating Analysis of Amazon Product Categories

Project Overview – Rating & Quality Signal Analysis

This phase of the project analyzed customer ratings across product categories and sub_categories to understand how well products meet customer expectations and where operational improvements are required.

The analysis compared average ratings at the sub_category level to identify stable, high trust segments versus categories showing consistent customer dissatisfaction. High rated categories were treated as low-risk, stable markets with strong product market fit, while lower-rated categories were flagged as quality or experience issues rather than marketing problems.

To go deeper, rating gaps within main categories were examined to separate structural quality failures from minor experience mismatches. Large gaps indicated fundamental issues such as material quality or supplier reliability, requiring operational fixes and quality audits. Smaller gaps pointed to problems in product presentation or expectation setting, where simple improvements in listings, images, or descriptions could lift ratings. Overall, this analysis reframed ratings as a diagnostic signal helping distinguish where growth can be driven through marketing versus where the product or supply chain must be fixed first.

Data Set Summary

Dataset Context

The dataset, Amazon Sales Data 2023, contains detailed information on 551,585 products across 19 main categories and 112 sub-categories.

It includes 9 columns covering product details, pricing, customer ratings, and access information.

Key Features

- Product details: Name, Main Category, Sub-category
- Pricing: Discount Price, Actual Price
- Ratings: Average Rating, Number of Ratings
- Media & access: Image URL, Product Link

This dataset supports large-scale marketplace analysis across multiple product categories.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Loaded the CSV file into a DataFrame and previewed the first 15 records to understand the data layout and contents.

	name	main_category	sub_category	image	link	ratings	no_of_ratings	discount_price	actual_price
0	Lloyd 1.5 Ton 3 Star Inverter Split AC (5 In 1...	appliances	Air Conditioners	amazon.com/images/I/31UISB90sY...	https://www.amazon.in/Lloyd-Inverter-Convertible...	4.2	2255.0	32999.0	58990.0
1	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (C...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Convertible-Anti-Viru...	4.2	2948.0	46490.0	75990.0
2	LG 1 Ton 4 Star AI DUAL Inverter Split AC (Copp...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Inverter-Convertible...	4.2	1206.0	34490.0	61990.0
3	LG 1.5 Ton 3 Star AI DUAL Inverter Split AC (C...	appliances	Air Conditioners	amazon.com/images/I/51Fb7FctD...	https://www.amazon.in/LG-Convertible-Anti-Viru...	4.0	69.0	37990.0	68990.0
4	Carrier 1.5 Ton 3 Star Inverter Split AC (Copp...	appliances	Air Conditioners	amazon.com/images/I/41lrtqXPW...	https://www.amazon.in/Carrier-Inverter-Split-C...	4.1	630.0	34490.0	67790.0

- **Standardized numeric fields** by removing currency symbols, commas, and text to ensure consistent numerical formatting.
- **Converted relevant columns to numeric data types** to make them suitable for calculations and analysis.
- **Removed duplicate products** using a defined business key to avoid double counting in analysis.
- **Renamed columns for consistency and clarity**, improving readability and downstream usability.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

Data Analysis using SQL (Business Transactions)

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Due to the length and complexity of the queries, the detailed SQL code is omitted, and a set of reference slides is shared to illustrate the structure and contents of the analysis, without implying a complete or sequential narrative.

